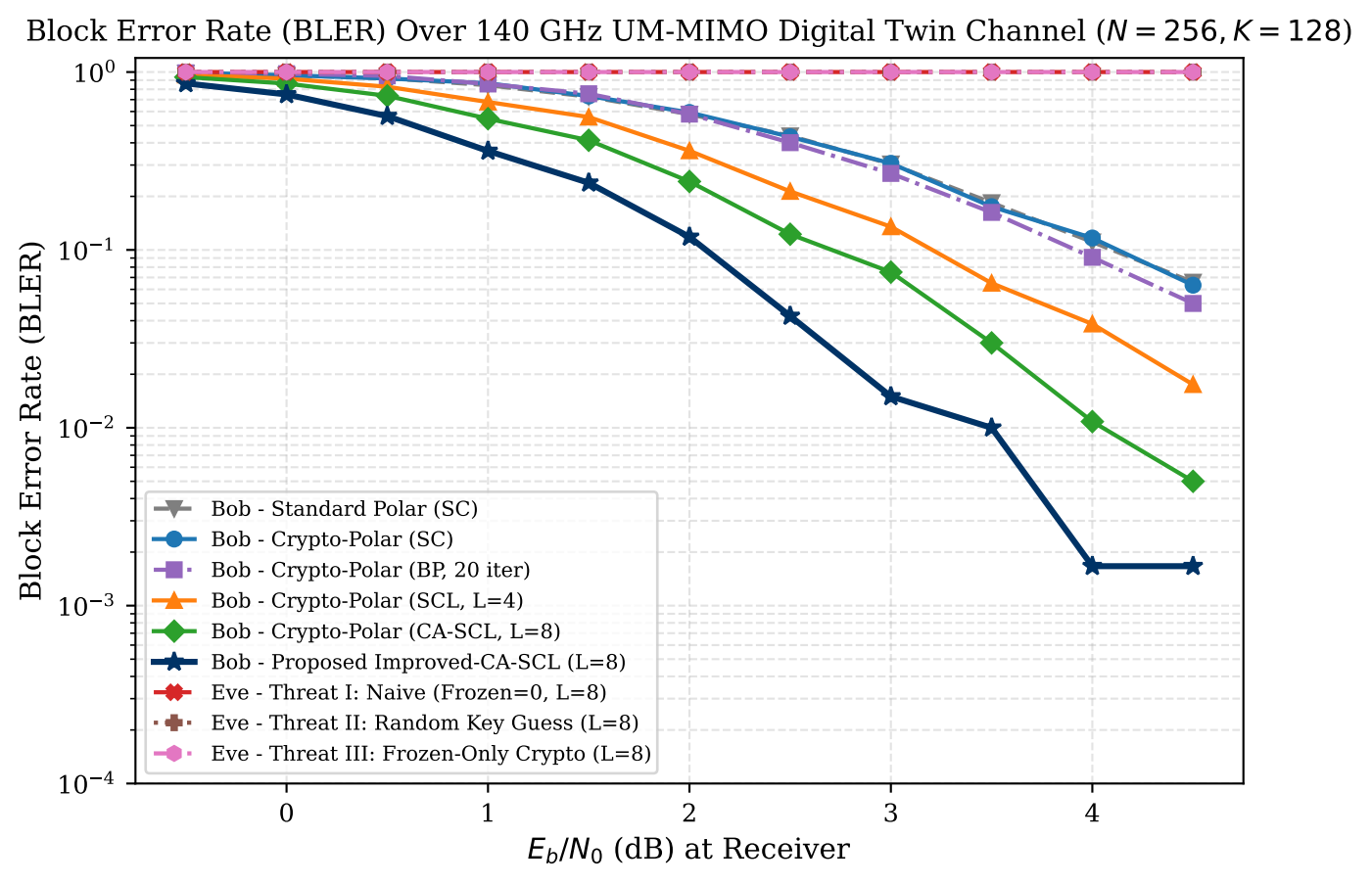
Block Error Rate (BLER) Over 140 GHz UM-MIMO Digital Twin Channel ($N = 256, K = 128$)
Block Error Rate (BLER)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
1
2
3
4
E_b/N_0 (dB) at Receiver
Bob - Standard Polar (SC)
Bob - Crypto-Polar (SC)
Bob - Crypto-Polar (BP, 20 iter)
Bob - Crypto-Polar (SCL, L=4)
Bob - Crypto-Polar (CA-SCL, L=8)
Bob - Proposed Improved-CA-SCL (L=8)
Eve - Threat I: Naive (Frozen=0, L=8)
Eve - Threat II: Random Key Guess (L=8)
Eve - Threat III: Frozen-Only Crypto (L=8)